5 CHAPTER 5: SYSTEM ANALYSIS AND DESIGN

5.1 Overview

This chapter presents the system analysis and design of the proposed USIU Campus Navigation System. It outlines the structural and functional models that define how the system operates, interacts with users, and achieves its intended objectives. The chapter introduces various diagrams, including system architecture, use case models, flowcharts, data flow diagrams, class diagrams, activity diagrams, and sequence diagrams, which collectively illustrate the internal workings of the system. Additionally, wireframes are provided to showcase the user interface design and navigation flow, offering a preview of the end-user experience. By integrating both conceptual and visual representations, this chapter establishes a clear foundation for the implementation phase, ensuring that the navigation system is both functional and user-friendly.

5.1 System Models

5.1.1 System Architecture Diagram

A system architecture diagram is a visual representation that shows how different components of a system interact and work together. It illustrates the structure, data flow, and relationships between the user interface, application logic, and data storage layers. In the case of the Campus Navigation System, the diagram outlines how users, the mobile application, backend server, and external services like Google Maps connect to deliver location and navigation features efficiently.

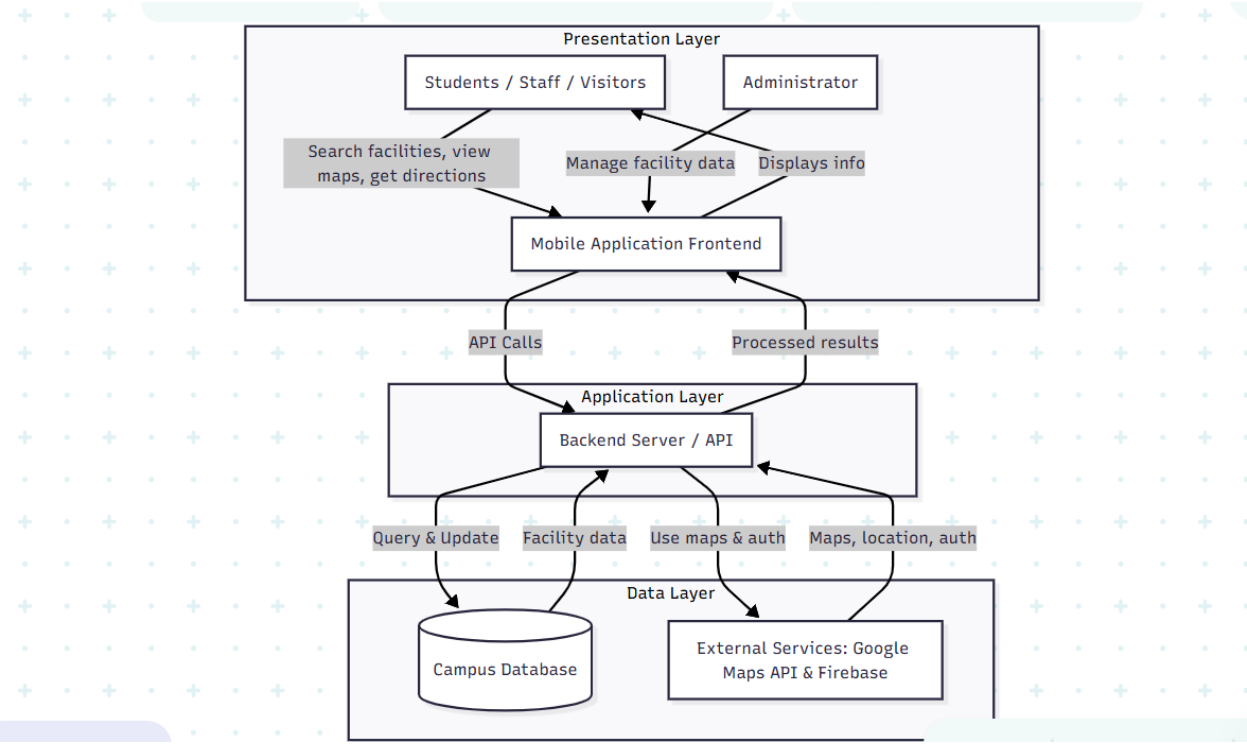

Figure 12: System Architecture Diagram

Figure 12 above shows the architecture of the Campus Navigation System follows a three-tier design consisting of the Presentation Layer, Application Layer, and Data Layer. Each layer performs specific roles that collectively enable the system to function efficiently, ensure modularity, and support real-time interactions between users and system services.

1. Presentation Layer

This layer represents the user interface and interaction point of the system. It consists of Students, Staff, Visitors, and Administrators, who access the system through a Mobile Application Frontend.

Students, Staff, and Visitors can search for campus facilities, view maps, and get directions to classrooms, offices, or other locations within the institution.

The Administrator manages facility data such as building names, locations, and categories, ensuring the information displayed to users is accurate and up to date.

The frontend communicates user requests (such as searching for a location or requesting directions) to the backend via API calls and receives processed results to display as interactive maps or location details.

This layer ensures that users interact with an intuitive, responsive, and user-friendly interface, providing a seamless navigation experience.

2. Application Layer

The Application Layer acts as the intermediary between the user interface and the underlying data resources. It is primarily powered by a Backend Server / API, which handles business logic, data processing, and communication between the frontend and database.

- The backend processes incoming API requests from the mobile application, such as location searches or facility management updates.
- It integrates with external services (like Google Maps API) to process geolocation data, route generation, and map visualization.
- Once data is processed, the backend sends the results back to the frontend for user display.

This layer ensures real-time functionality, secure data exchange, and smooth coordination between system components.

3. Data Layer

The Data Layer is responsible for managing and storing all essential information used by the system. It consists of:

- A Campus Database, which stores structured information such as building details, facility locations, user profiles, and navigation data. The backend queries and updates this database whenever users perform searches or administrators modify facility information.
- External Services such as Google Maps API and Firebase, which support advanced mapping, location tracking, authentication, and real-time data synchronization.

This layer ensures that the system operates with reliable data storage, accurate geolocation services, and secure access control.

5.1.2 Use Case Diagram

A Use Case Diagram is employed to model the interactions between users (actors) and the system, illustrating the different services the application will provide. For the USIU-A campus navigation system, the diagram identifies the primary actors and the use cases they are associated with. This ensures that the system functionalities remain user-centered and aligned with the needs established in the requirements phase.

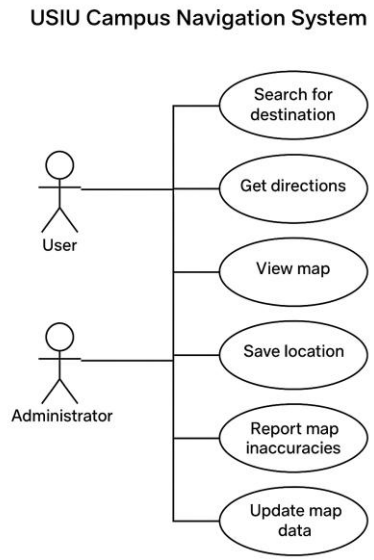

Figure 13:Use Case Diagram

Figure 13 above shows the use case diagram and it illustrates the key interactions between users and the Campus Navigation System. It identifies four primary actors: Student/Visitor, Admin, System (Automated Functions), and Maintenance Personnel.

The Student/Visitor interacts with the system to search for specific campus locations such as lecture halls, offices, cafeterias, and hostels. They can also view real-time navigation routes, access building details, check event or service locations, and receive notifications on temporary closures or route changes.

The Admin is responsible for managing the system database, which includes updating campus maps, adding or modifying location data, handling user accounts, and monitoring system performance. The admin also ensures that all navigation information remains accurate and up to date.

The Maintenance Personnel work closely with the admin to report or update information related to changes on the campus infrastructure—such as building renovations, new constructions, or blocked paths—to keep the navigation data reliable and current.

Meanwhile, the System (Automated Functions) performs background tasks such as providing route suggestions based on the user's current location, updating map data, sending notifications for map changes, and generating analytical reports on user activity.

The diagram also demonstrates various relationships such as «include» for shared processes like user authentication (login/sign-up) and location search, and «extend» for optional or conditional processes like route recalculation, event notifications, or dynamic updates triggered by real-time campus changes.

5.1.3 Entity Relationship Diagram (ERD)

Introduction

An ERD illustrates the logical structure of the database by showing entities, their attributes, and the relationships between them. For the USIU Campus Navigation System, the ERD is essential because the system relies heavily on a database to store and retrieve information about buildings, facilities, users, and navigation data.

This ensures that the system can quickly return accurate results when students, staff, or visitors search for a building, request directions, or access facility details.

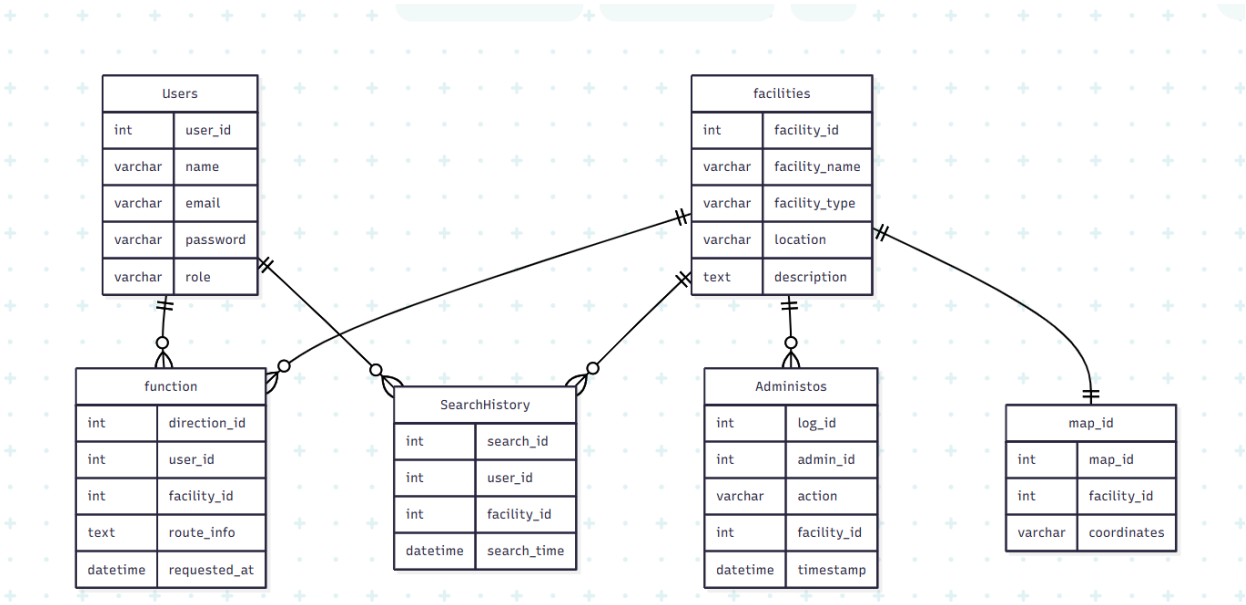

Figure 14: Entity Relationship Diagram (ERD)

Figure 14 above shows the Entity-Relationship Diagram (ERD) for the Campus Navigation System , it illustrates the key entities within the system and how they relate to each other. It defines the structure of the database by showing entities such as Users, Facilities, Locations, and Navigation Routes, along with their attributes and connections.

For instance, a *User* (which could be a student, staff, or visitor) can search or access multiple *Facilities* on campus, while each *Facility* is linked to a specific *Location*. The *Routes* entity helps store path data between locations to enable navigation and direction features.

This diagram ensures efficient data organization and supports the interaction between the mobile application and backend database, enabling smooth retrieval, storage, and management of campus-related information.

5.1.4 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a visual representation of how data moves within a system. It focuses on the flow of information between processes, data stores, and external entities, showing how inputs are transformed into outputs through different system functions. Unlike flowcharts that show the sequence of operations, a DFD emphasizes what data is being transferred and where it flows, helping to understand the logical structure of a system. It is typically developed in levels—

starting from a general overview (Level 0) and moving to more detailed diagrams (Level 1, Level 2, etc.) that break down specific processes.

i. Level 0: Context Diagram

The Level 0 DFD, also known as the Context Diagram, represents the system as a single process and shows how it interacts with external entities. It provides a high-level view of data flow without going into detailed processes.

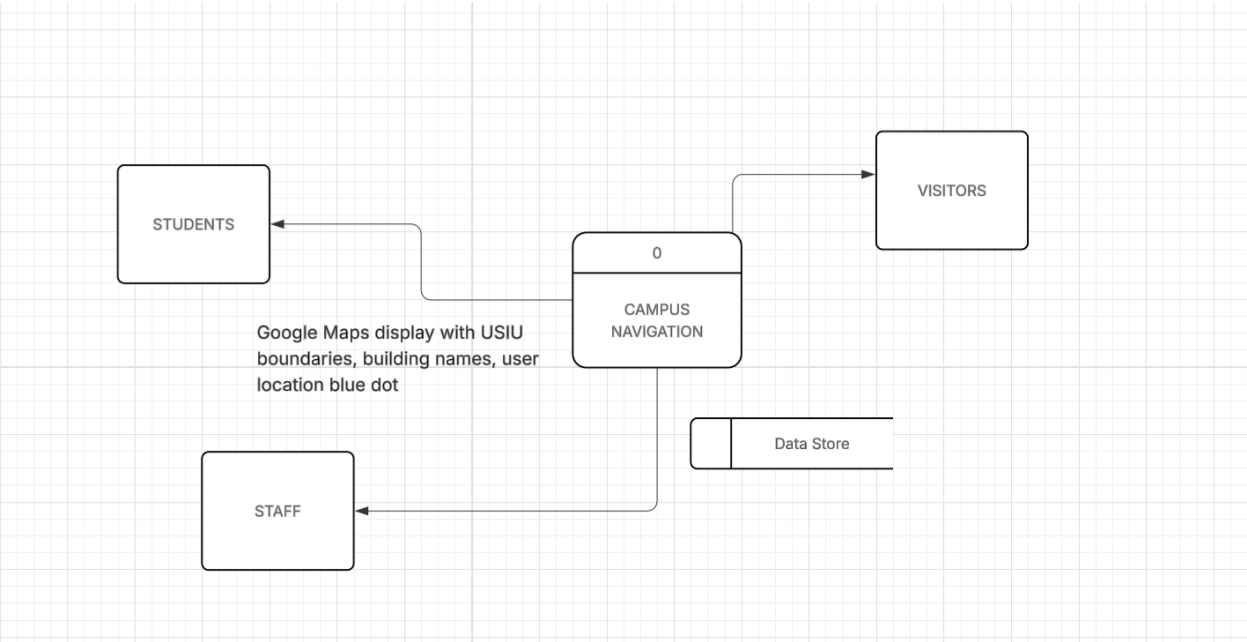

Figure 15: Level 0 Context Diagram

Figure 15 above shows a level 0 context diagram, users serve as the main external entities who provide input to the system through location search queries, route requests, or specific building lookups. For example, a student may search for the quickest route to a lecture hall, or a visitor may request directions to administrative offices or parking areas. These requests are transmitted to the Campus Navigation System, which processes the input and returns corresponding navigation results such as route maps, estimated walking times, or alternative paths in case of restricted areas.

The system also interacts with administrators, who provide data inputs such as updated building locations, facility information, or notices about inaccessible areas. This ensures that users receive

accurate and up-to-date navigation results. The administrator's role helps maintain the reliability and accuracy of the data used by the system.

Essentially, the Level 0 DFD focuses on depicting the exchange of data between users and the system without delving into how each function is executed internally. It shows the overall data movement, where external users send queries to the Campus Navigation System, and the system, in turn, outputs relevant route details and location information. This provides a clear and simplified visualization of the system's scope, emphasizing the input and output data flows that define the system's primary purpose — to help users navigate the university campus easily and efficiently.

ii. Level 1: Detailed DFD

The Level 1 Data Flow Diagram (DFD) of the Campus Navigation System provides a more detailed representation of how the system functions internally by decomposing the single process shown in the Level 0 diagram into several interconnected sub-processes. This breakdown helps in understanding the logical flow of data within the system and how different components interact to deliver navigation services to users. Unlike the Level 0 DFD, which only shows external interactions, the Level 1 diagram dives deeper to explain the internal data transformations and the relationships between various processes, data stores, and external entities.

In this level, the main system process is divided into smaller, well-defined sub-processes such as User Authentication, Search Location, Generate Navigation Route, and Update Campus Data. Each of these processes performs a specific function that contributes to the overall efficiency and reliability of the Campus Navigation System. For instance, the *User Authentication* process ensures that only authorized users, such as students and staff, can access certain system features. The *Search Location* process allows users to input queries, such as searching for a particular classroom or facility, while the *Generate Navigation Route* process uses that input to determine the most optimal path within the campus. Meanwhile, Update Campus *Data* enables administrators to add or modify information such as new buildings, blocked routes, or event locations.

The diagram also highlights the data stores that support these processes primarily the Campus Location Database and the User Information Repository. These repositories store essential information that allows the system to respond accurately and quickly to user requests. Data

continuously flows between the users, the system processes, and these storage components, ensuring up-to-date navigation results and reliable user experiences.

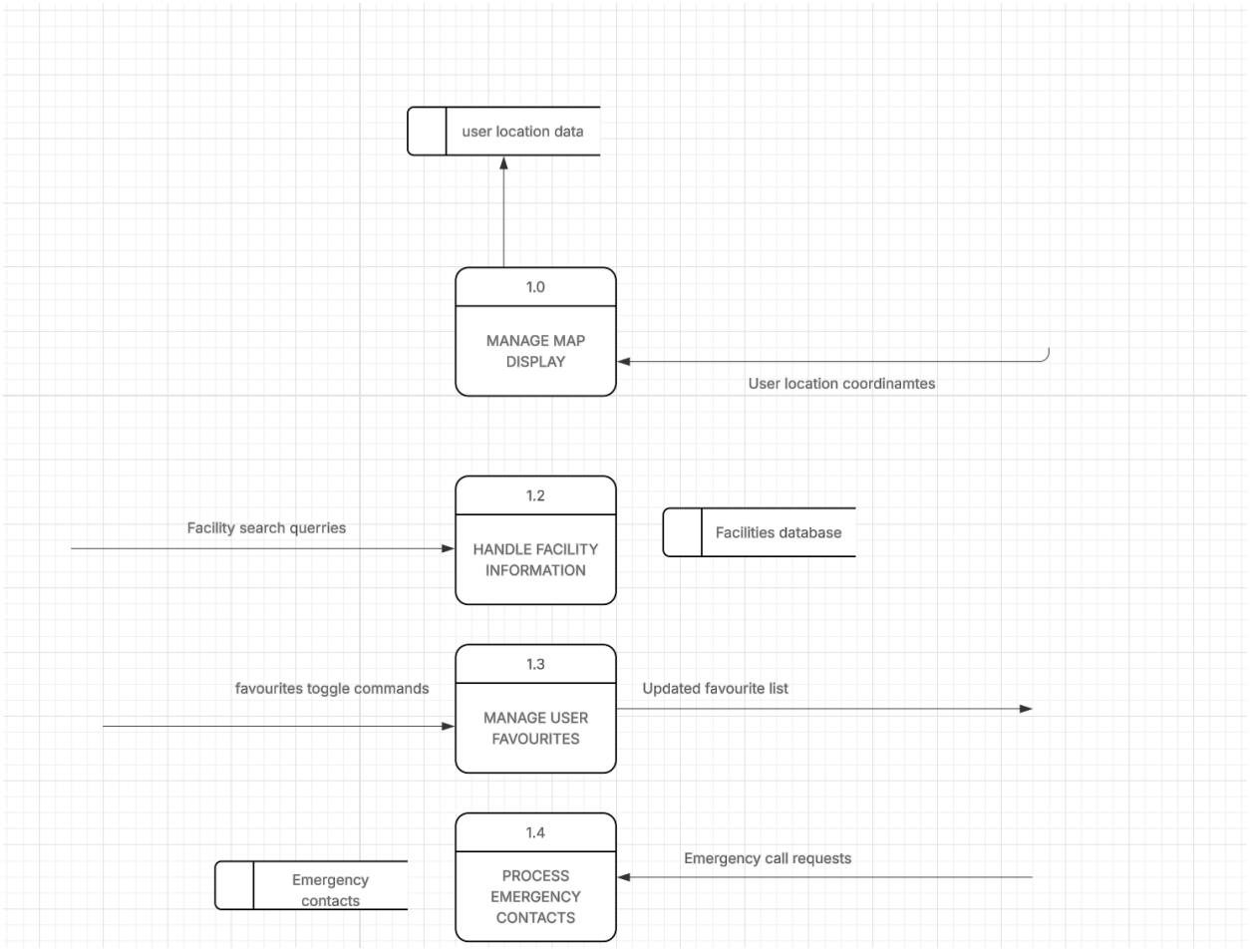

Figure 16: Level 1: Detailed DFD

Figure 16 above shows the Level 1 DFD further that emphasizes how these interconnected processes collaborate to produce smooth and effective system functionality. It demonstrates the logical pathways that data follows — for example, when a user searches for a lecture hall, the system retrieves relevant information from the Campus Location Database, processes it through the Generate Navigation Route function, and returns detailed directions back to the user interface.

Similarly, when administrators update or modify campus data, the new information is stored and made immediately available to users for subsequent navigation requests.

This level of detail provides valuable insight into how the system's internal components support each other to maintain accuracy and real-time responsiveness. It showcases the efficiency of data flow within the Campus Navigation System, ensuring that the right information is delivered to the right user at the right time. The Level 1 DFD therefore serves as a bridge between the broad overview provided in Level 0 and the more granular processes detailed in Level 2, illustrating a well-structured and logically connected system that facilitates seamless movement and accessibility across the campus.

iii. Level 2 : Detailed DFD

The Level 2 Data Flow Diagram for the Campus Navigation System provides a detailed breakdown of the major processes that were introduced in the Level 1 DFD. While Level 1 focused on the broader interaction between the user, the system, and the main processes such as "User Authentication," "Location Search," and "Navigation Display," this Level 2 diagram expands those processes to show exactly how data moves within each function. It helps visualize how information is entered, processed, stored, and output in more specific detail.

The Level 2 DFD starts by examining the User Authentication process in depth. When a user logs in, their credentials are captured by the Input User Data sub-process. The information is then passed to the Validate Credentials sub-process, which checks it against the User Database**.** If the credentials match, the system allows access through the Grant Access sub-process and directs the user to the home dashboard. If not, an error message is sent back to the user interface for correction.

The next major process is the Search Location function. At this level, it is broken down into smaller steps such as Receive Search Query**,** Verify Search Input**,** Fetch Location Details**,** and Display Search Results**.** When the user types a name—like "Library" or "Administration Block"—the system first validates whether the name exists in the Location Database. Once verified, the relevant coordinates and building details are retrieved and displayed on the screen.

Another core process in this level is Generate Route. This part is expanded into Capture Starting Point, Retrieve Destination Coordinates, Calculate Optimal Path, and Display Navigation Route.

The sub-processes show how the system determines the user's current location, retrieves the destination's coordinates from the Map Database, computes the shortest or most convenient route, and then visually presents it on the app's interactive map.

The Level 2 DFD also includes background processes like Update Map Information and Manage Feedback. The update function allows the system administrator to refresh location data or add new facilities through the Admin Panel, which is linked to the Location Database. The feedback process collects user input from the Feedback Form, processes it under Store Feedback Data, and saves it to the Feedback Repository for later analysis or system improvement.

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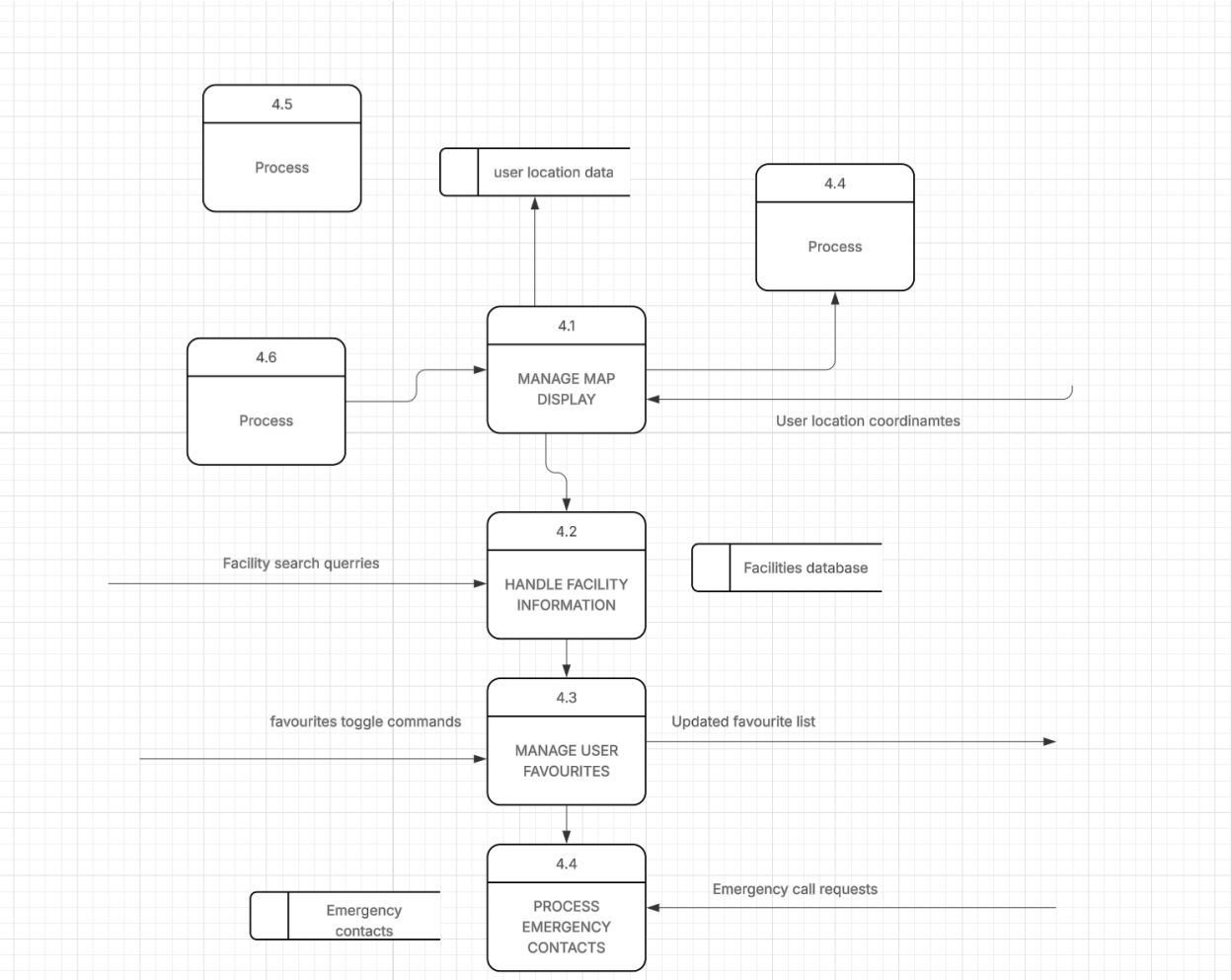

Figure 17:Level 2 : Detailed DFD

Figure 17 above shows the level 2 DFD diagram.After developing the Level 2 DFD, the data flow of the entire Campus Navigation System became clearer and more transparent. Each interaction between the user, databases, and internal processes could now be traced step by step. For instance, when a student logs in and searches for a lecture hall, the diagram clearly shows how the request moves through authentication, validation, database retrieval, and finally to map display. This detailed flow ensures that no critical data movement is overlooked during implementation.

The diagram also highlights how internal components are interlinked. The User Database, Location Database, and Map Database interact constantly, supporting the smooth exchange of data. This interconnected structure ensures that users always receive up-to-date navigation details and accurate building information. Additionally, the flow from Generate Route back to the Display

Results process ensures that users can see both their current position and the recommended route on one screen, improving usability.

After the diagram was completed, it became possible to analyze the efficiency of data handling within the system. The team could identify potential issues such as duplicated data paths or unnecessary processes and optimize them before coding. The Level 2 DFD therefore acted as a problem-prevention tool, reducing the risk of system errors during development. It also helped in finalizing how the databases would be structured and how the app's backend logic would interact with the user interface.

In conclusion, the Level 2 DFD for the Campus Navigation System provides a comprehensive view of the system's data movement and logic. It breaks down each main process—user login, location search, route generation, and feedback management—into clear, traceable sub-processes. Before the diagram, planning focused on identifying key system activities and their relationships; after the diagram, these ideas were refined into a working data structure ready for development. The Level 2 DFD ultimately serves as a bridge between conceptual design and actual programming, ensuring that every function of the app operates efficiently and consistently.

5.1.5 Flowchart

A flowchart is a visual representation that outlines the step-by-step flow of processes or activities within a system. It uses standardized symbols such as ovals, rectangles, diamonds, and arrows to depict actions, decisions, and the direction of process flow. In the context of the Campus Navigation System, the flowchart provides a clear and logical view of how users interact with the system from start to finish. It visually demonstrates the sequence of operations — from user login and input of location queries to the generation of navigation routes and display of results. This makes it easier to understand the logical order of events, identify decision points, and visualize how the system responds under different user actions or conditions.

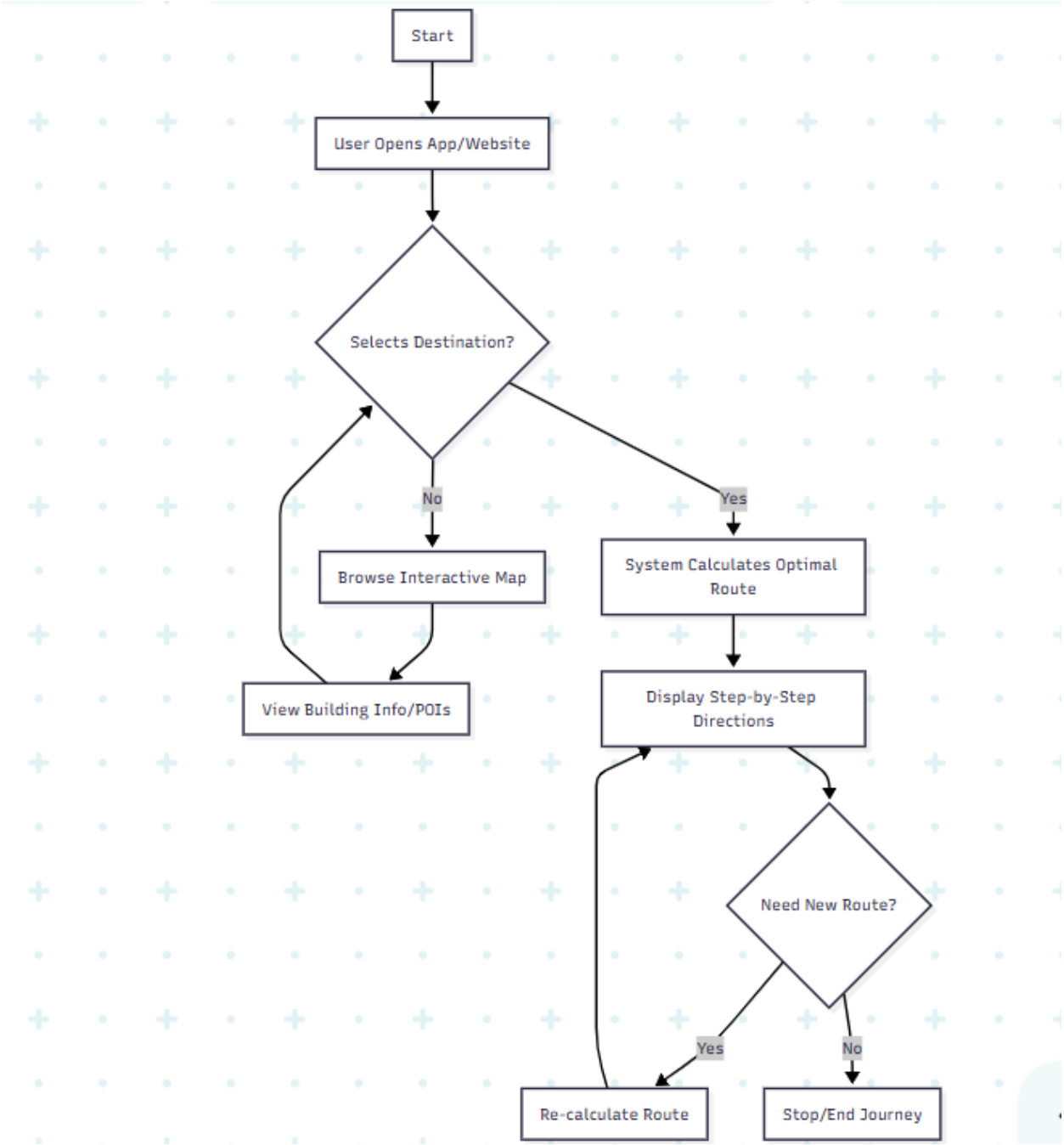

Figure 18:Flowchart

Figure 18 shows the flowchart of the Campus Navigation System that illustrates the systematic process a user follows when using the application. It begins when the user launches the system and logs in, after which the system verifies the credentials. If the login is successful, the user can

proceed to search for a specific location or building within the campus. Once the search query is entered, the system checks the Campus Location Database for a match. If the location exists, the system generates and displays the most efficient navigation route. However, if the location is not found, an error message is shown prompting the user to refine their search.

Additionally, the flowchart demonstrates how administrators and security officers interact with the system to update or modify campus information. The process ensures that new data is stored and reflected in real-time for all users. By visually presenting each step, the flowchart helps to clarify the logical flow of operations, simplifies system understanding, and supports the development and troubleshooting processes. It effectively captures how different users navigate through the system, making it a valuable tool for system design, documentation, and future improvements.

5.1.6 Activity Diagram

An activity diagram is a type of behavioral diagram used in system modeling to represent the flow of activities and actions within a process. It focuses on how different tasks are carried out in sequence, the decisions made along the way, and how various activities connect to achieve a specific goal. In the context of the Campus Navigation System, the activity diagram visually demonstrates the step-by-step workflow from when a user interacts with the system to when they receive navigation results or updates. It helps illustrate both user-driven and system-driven processes, such as logging in, searching for locations, generating routes, and updating campus data. By using decision points, parallel activities, and start/end states, the activity diagram gives a clear view of the logical flow of operations and how different parts of the system collaborate to deliver smooth navigation and user support.

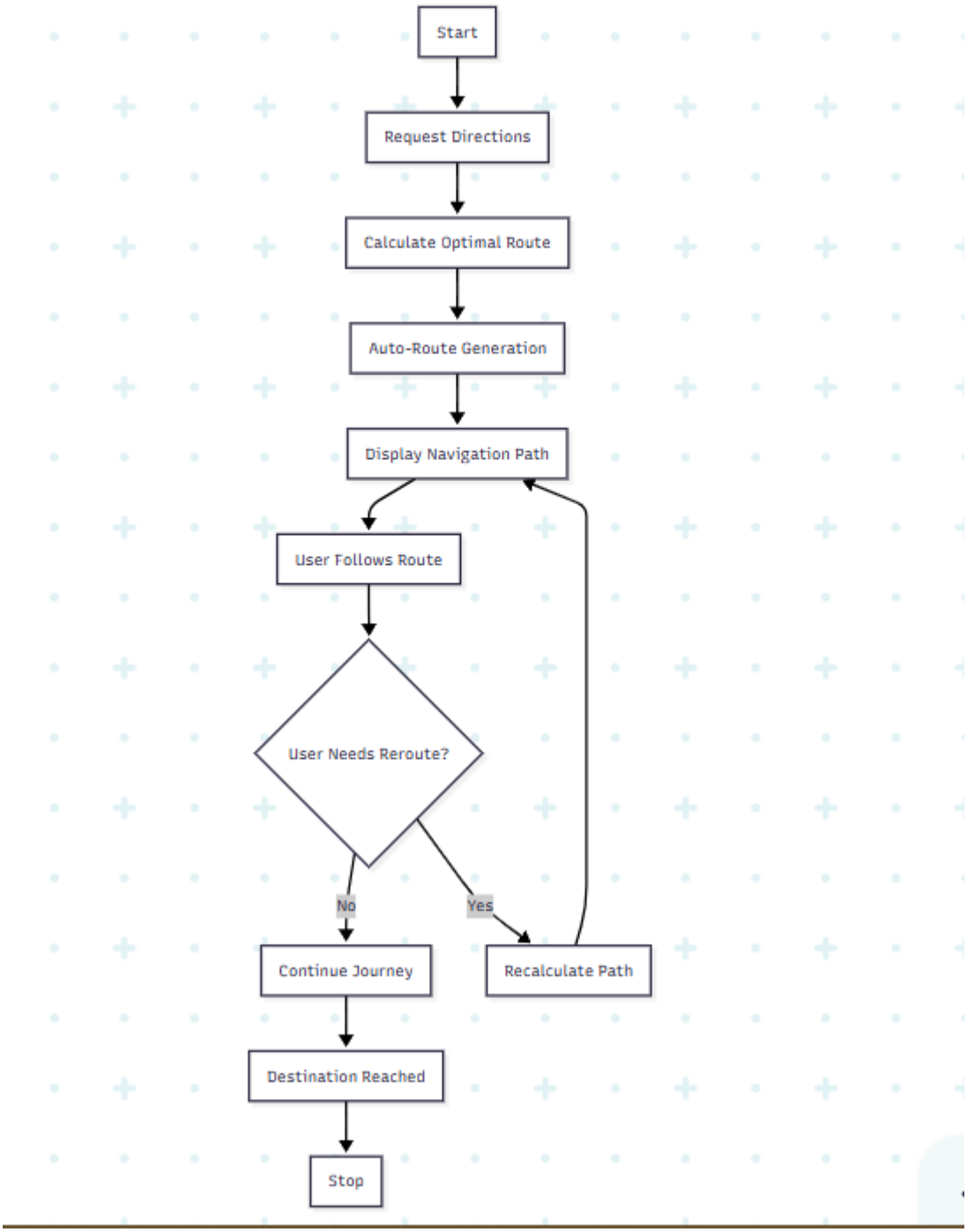

Figure 19:Activity Diagram

Figure 19 shows the activity diagram for the Campus Navigation System that outlines the logical sequence of actions that occur when users interact with the platform. The process begins with the

user login or registration, after which the system verifies credentials. Once authenticated, the user can perform actions such as searching for a location, viewing campus routes, or requesting navigation assistance. The diagram shows how these actions flow from one stage to another — for instance, after entering a search query, the system retrieves relevant location data, processes it through the navigation logic, and displays the optimal route.

For administrative users, the activity diagram includes additional actions such as adding or updating campus locations and managing user data. Decision nodes are used to represent choices, such as whether the entered location exists in the database or whether an update is required. If conditions are met, the process continues; otherwise, the system prompts the user to try again or provides alternative options.

Overall, the activity diagram provides a detailed visualization of how the Campus Navigation System handles both normal and exceptional scenarios. It emphasizes the smooth flow of data and decisions from start to finish, ensuring efficient user interactions and accurate delivery of navigation information across different user roles within the system.

5.1.7 Sequence diagram

A sequence diagram is a type of interaction diagram used in system design to show how objects and components interact with each other over time. It focuses on the order of messages exchanged between different entities in a system to accomplish a specific function or process. Sequence diagrams are particularly useful in visualizing dynamic behaviors such as user actions, system responses, and data exchanges. In the Campus Navigation System, the sequence diagram demonstrates how users — including students, visitors, and administrators — interact with the system to perform tasks like searching for locations, generating navigation routes, and updating campus data. It helps designers and developers understand the timing and flow of operations, ensuring that each process occurs in a logical and efficient order.

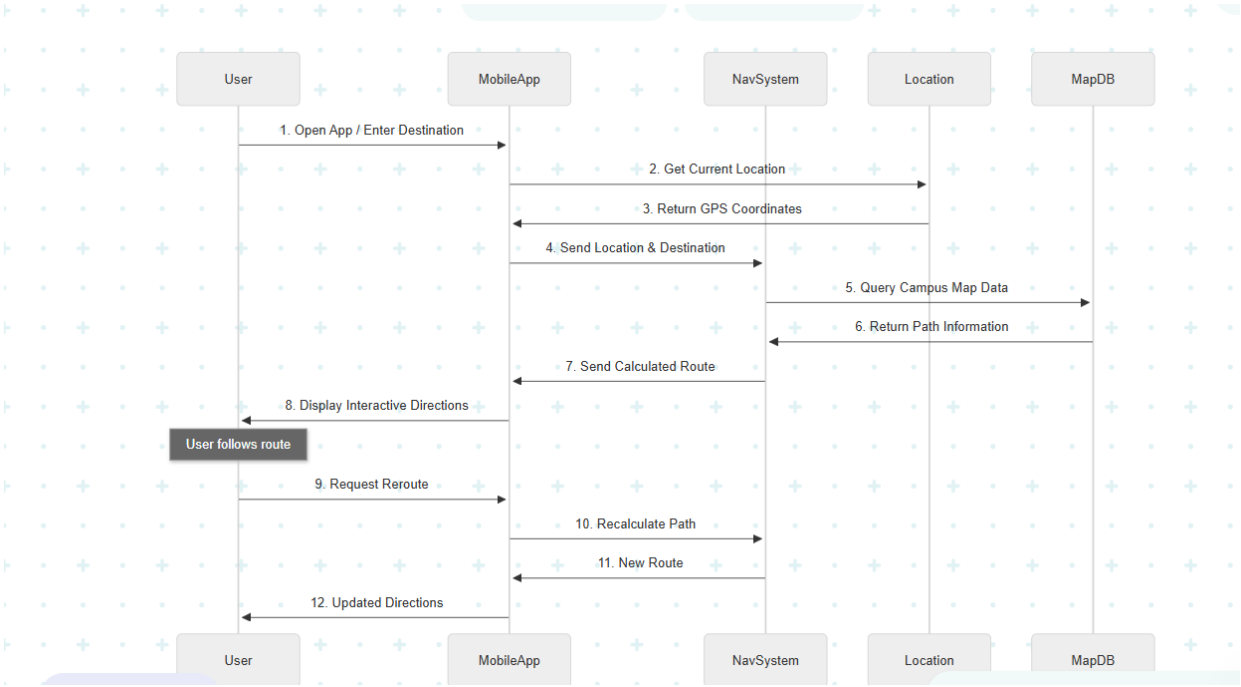

Figure 20: Sequence diagram

Figure 20 shows the sequence diagram for the Campus Navigation System that illustrates the step-by-step flow of interactions between key components, including the User, System Interface, Navigation Controller, Database, and Admin. The process begins when a User initiates an action such as searching for a specific campus location or requesting navigation assistance. The System Interface captures this request and forwards it to the Navigation Controller, which handles the main logic of the operation. The Navigation Controller then interacts with the Database to retrieve relevant data for example, location details, building coordinates, or stored routes.

Once the data is retrieved, the Navigation Controller processes the information and generates the required navigation route, which is then sent back through the System Interface for display to the User. For administrative tasks, the Admin actor may initiate operations like adding new locations or updating existing campus details, which follow a similar interaction pattern involving data verification and database updates.

Overall, the sequence diagram captures the real-time interaction between users and the system, clearly showing how requests flow through various components to produce accurate and timely

responses. It ensures that every function — from user login to navigation route generation — is executed in the correct sequence, promoting system reliability and user satisfaction.

5.1.8 Class diagram

A class diagram is a type of structural diagram in system design that illustrates the blueprint of an object-oriented system by showing the system's classes, their attributes, methods, and the relationships among them. It provides a static view of the system, helping developers understand how different components interact and depend on one another. In the Campus Navigation System, the class diagram serves as the foundation for understanding how data and functionality are organized within the system. It helps in identifying key entities such as users, locations, routes, and administrators, along with their specific roles and interactions. This visualization ensures that the system's architecture is well-structured, supports reusability, and maintains clear relationships between the different parts of the system.

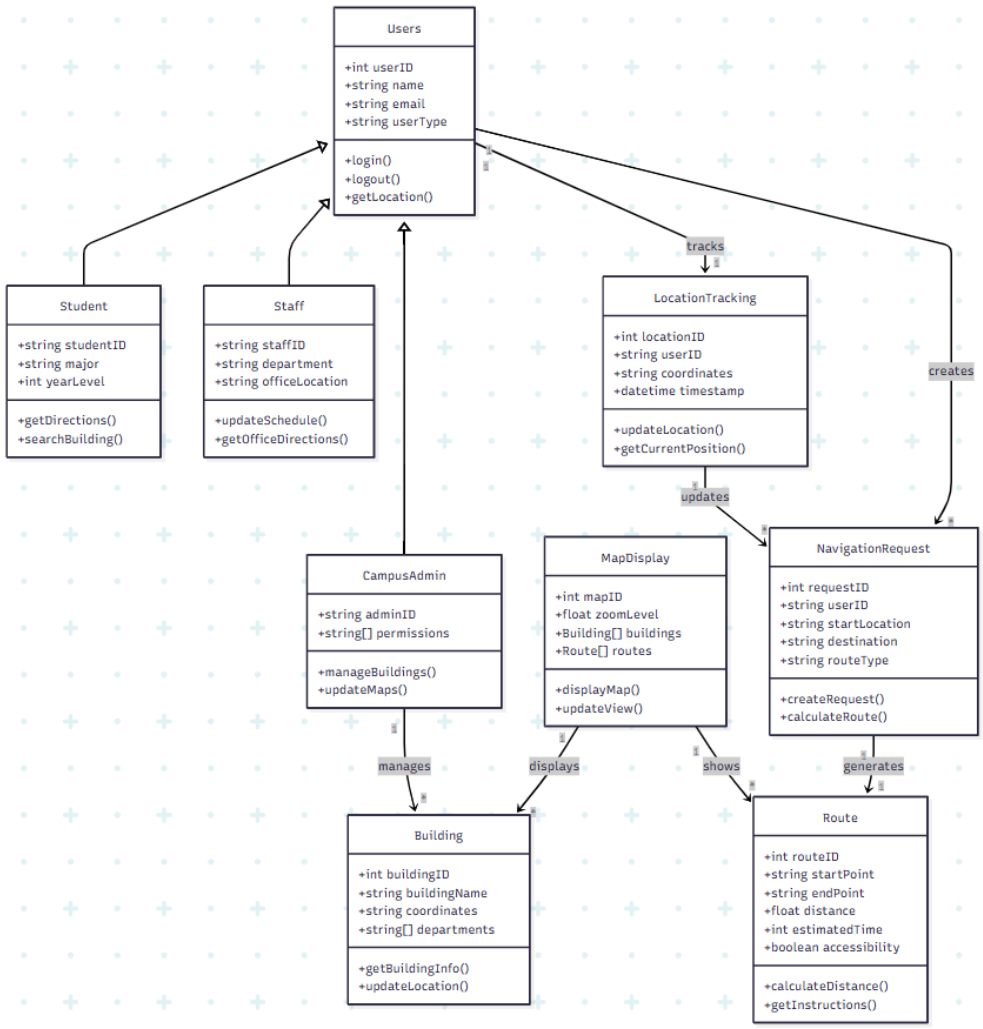

Figure 21: Class diagram

Figure 21 shows the class diagram for the Campus Navigation System that depicts the main classes and their interconnections that support the system's functionality. Key classes include User, Admin, Location, Route, and NavigationSystem. The User class represents all users of the system, such as students and visitors, and includes attributes like user ID, name, and login credentials. The Admin class extends the User class, adding additional responsibilities like updating campus data, managing locations, and overseeing the system's operation.

The Location class stores information about various campus places such as lecture halls, offices, and facilities, while the Route class manages navigation details, including start and end points and

the path generated between them. The NavigationSystem class serves as the central component that coordinates user requests, retrieves data from the Location and Route classes, and provides users with accurate navigation instructions. Relationships such as associations and dependencies are clearly shown — for example, a User interacts with the NavigationSystem to search for a location, and the system, in turn, communicates with the Route class to generate a navigation path.

Overall, the class diagram illustrates how different entities within the Campus Navigation System work together to support efficient campus guidance, data management, and user interaction, forming the backbone of the system's design and implementation.

5.1.9 Wireframe

The wireframes for the Campus Navigation System serve as the main blueprint of the entire mobile app. They outline how each screen will look, what components will appear, and how users will move from one function to another. At this stage, the focus is not on colors or decoration but on structure and functionality. The wireframes help to ensure that features like search, navigation, location markers, and campus details are well-organized before actual coding begins.

Wireframing played a key role in transforming the idea of the app into a structured plan. Since the app aims to help students, staff, and visitors find their way around campus, it was important to visualize how users would move between pages like the login page, home screen, search bar, and map interface. The wireframes make it easy to see how each of these sections will connect and how users can quickly locate buildings, departments, or routes.

Before creating the diagrams, the main user journeys were mapped out in simple lists and flowcharts. For example, a user might open the app, sign in, tap the search bar, type the name of a lecture hall, and then view the route to reach it. These steps helped identify what pages were necessary and what elements needed to be included on each screen.

The initial wireframes were low-fidelity, meaning they were drawn using simple boxes, circles, and labels. These rough sketches helped to decide where to place major elements like navigation menus, icons, and buttons. For instance, the top of the screen might include a search field, the center area might display the interactive map, and the bottom part might hold shortcut buttons for quick access to key features. This stage helped create a clear structure that both designers and developers could understand.

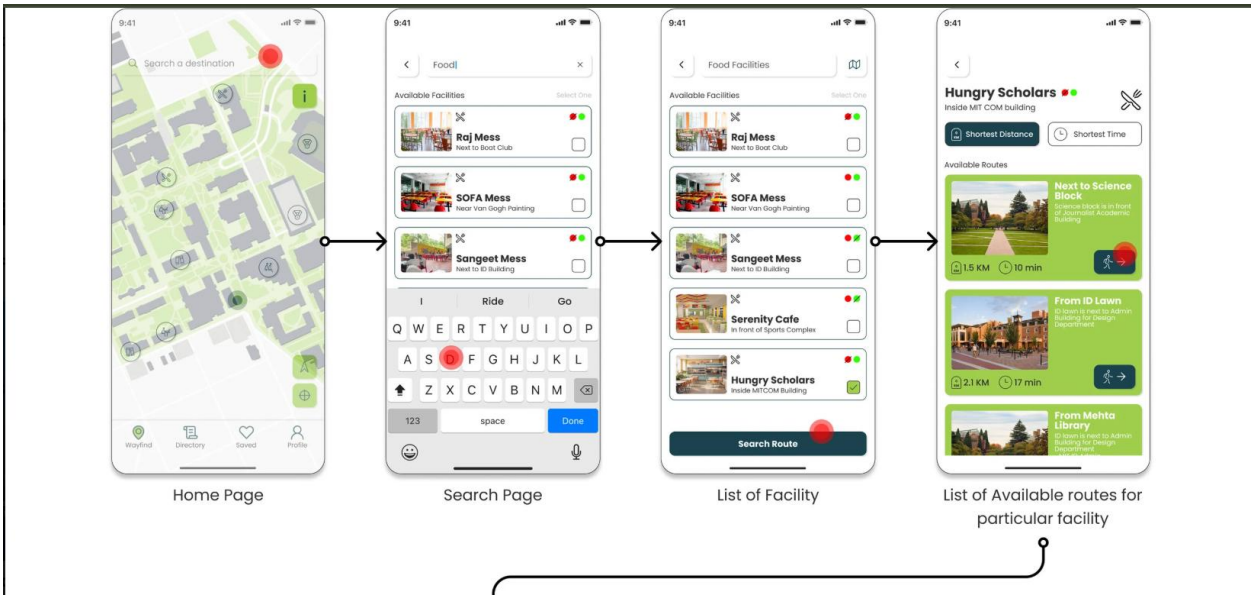

Figure 22: Wireframe

Figure 22 above represents the wireframe for the campus navigation system. The first page of the USIU Campus Navigation System wireframe focuses on the maps and search functionality, which serves as the central point of interaction for users. A prominently placed search bar allows students, staff, and visitors to type in the name of a building, department, or facility, with an auto-complete feature that suggests relevant locations as they type, thereby improving efficiency and reducing search time. The main area of the page is dedicated to an interactive digital map of the USIU-Africa campus, which allows zooming, panning, and selection of specific locations. When a user selects a facility, a pin marker appears on the map along with a pop-up card displaying details such as the name, function, and distance from the user's current location. Integration of GPS-based real-time tracking enables users to see their live position and follow a visual route toward their destination.

Additional features on this page include filters and layer controls that allow users to toggle between different categories of locations, such as academic buildings, libraries, cafeterias, or hostels. This design choice ensures that users can customize the map view according to their specific needs, making the system more versatile and user-centered. The interface emphasizes clarity and simplicity, ensuring that even first-time visitors can navigate the campus effectively while reducing cognitive load through intuitive design elements. Overall, this page establishes the foundation for

a seamless navigation experience by combining interactive mapping, real-time location tracking, and user-friendly search functionality.

The second page presents different school facilities in a structured and organized layout. Facilities are grouped into categories such as academic buildings, administrative offices, student services, and recreational areas. Each facility is represented by a clickable icon or card that provides key information, including the facility name, purpose, and any operational details. This page enhances user exploration by allowing students and visitors to browse the campus visually rather than relying solely on search queries. Quick-access tabs or filters enable users to narrow down facilities by category, which improves usability and reduces the time needed to locate a desired destination. The design also emphasizes consistency in visual elements and labeling to ensure that users can quickly recognize different types of facilities at a glance.

The third page focuses on favorited facilities, offering users the ability to save frequently visited or important locations for quick access. Each favorited facility is displayed in a list or grid format with key details, such as distance from the user and a shortcut to navigation. Users can manage their favorites by adding or removing facilities, creating a personalized and efficient navigation experience. This feature supports daily campus routines and enhances convenience, especially for students who regularly move between lecture halls, libraries, and other frequently used spaces. Visual indicators, such as star icons or bookmarks, make the interface intuitive and allow users to identify their saved locations at a glance.

The fourth page is dedicated to emergency contacts, providing quick access to critical services such as campus security, medical assistance, and administrative support. Contact information is displayed prominently along with icons that represent each service, and a one-touch call or message feature is integrated to allow immediate communication in urgent situations. The design prioritizes accessibility, ensuring that users can locate emergency contacts quickly without navigating through multiple menus. Color coding, consistent iconography, and clear labeling enhance visibility and reduce response time during emergencies. This page emphasizes safety and preparedness, reinforcing the system's role as not only a navigation tool but also a resource for campus security and student wellbeing.

5.2 Functional Requirements

Functional requirements define the specific actions and features that the Campus Navigation System must provide to meet the needs of students, staff, and visitors.

Student Functional Requirements:

- FR1: The system shall allow students to search for campus buildings, lecture halls, offices, and amenities (e.g., cafeteria, library, hostels).
- FR2: The system shall provide step-by-step navigation to guide students from their current location to a selected destination.
- FR3: The system shall allow students to filter destinations by category (e.g., academic, administration, social, or residential).
- FR4: The system shall allow students to view building details (room codes, department offices, operating hours).
- FR5: The system shall provide notifications or prompts when they are near their selected destination.

Staff Functional Requirements:

- FR6: The system shall allow staff members to update or verify campus building details (e.g., office location changes).
- FR7: The system shall allow staff to search and navigate to offices, classrooms, or student service areas.
- FR8: The system shall allow authorized staff to add or update administrative locations.

Visitor Functional Requirements:

- FR9: The system shall allow visitors to use a simplified mode to search for common destinations (e.g., admissions office, auditorium, parking, clinic).
- FR10: The system shall provide route guidance optimized for new users unfamiliar with campus layout.

Administrator Functional Requirements:

- FR11: The system shall allow administrators to manage building data, routes, and geolocation records in the database.
- FR12: The system shall allow administrators to approve or edit map changes before they go live.
- FR13: The system shall generate usage reports (e.g., most-searched buildings, peak navigation times).

System-Wide Functional Requirements:

- FR14: The system shall provide real-time GPS navigation with step-by-step directions.
- FR15: The system shall allow search and filter functionality for browsing destinations by name, code, or category.
- FR16: The system shall ensure secure authentication for students, staff, and administrators.
- FR17: The system shall maintain a centralized geospatial database for campus building and route data.
- FR18: The system shall provide a mobile-first design, ensuring accessibility on both Android and iOS.

5.3 Non-Functional Requirements

Non-functional requirements describe the qualities and constraints of the system rather than the specific features.

- NFR1: Performance Requirements
  The system should be able to handle multiple students and staff navigating simultaneously without crashing. Routes should generate within 3–5 seconds on a stable internet connection.
- NFR2: Security Requirements
  All users (students, staff, administrators) must log in using a username/email and password or single sign-on (SSO) with university credentials. Passwords must be stored securely (hashed, not in plain text). Only administrators can update the database.

- NFR3: Usability Requirements
  The system must have a clean and simple interface that users can understand without training. Navigation should be intuitive with clear menus and search functionality. The system must be mobile-friendly since most users will access it on smartphones.

- NFR4: Reliability Requirements
  The system should be available at least 95% of the time during normal use. Saved campus data should never be lost and should remain consistent in the database even after crashes or restarts.

- NFR5: Scalability
  The system should support an increasing number of users, buildings, and routes as the campus grows. The database should be expandable to accommodate future updates (e.g., new departments, hostels, or facilities).

5.4 Conclusion

The Campus Navigation System is a project designed to address the real challenges of orientation and wayfinding within the United States International University (USIU). The proposal has outlined the problem of campus visitors and students struggling to locate buildings and facilities, which results in time wastage, confusion, and inconvenience. By leveraging mobile technology and mapping tools, this system aims to provide a reliable, interactive, and user-friendly solution that enhances navigation across the campus.

The document has presented the project's background, problem statement, aims and objectives, and the proposed methodology through well-structured phases including requirements gathering, system design, implementation, testing, evaluation, deployment, and documentation. Supporting diagrams such as the system architecture, use case diagram, ERD, and data flow diagrams illustrate the structure and functionality of the system. In addition, the proposal has also included a program of work, project schedule through a Gantt chart, and detailed resources required to carry out the project.

The expected outcome is a functional mobile application that will allow students, staff, and visitors to search for facilities, view an interactive campus map, get accurate directions, and access

essential facility details. The administrator module will ensure that the system remains up to date by allowing secure management of campus data.

In conclusion, the Campus Navigation System will not only improve user experience but also strengthen the university's image as a modern, student- and visitor-friendly institution. With careful planning, proper resource allocation, and systematic implementation, the project is both feasible and impactful within the given timeframe.